

Key Features:

- **Steady-State Growth:** The number of cells and nutrient levels remain nearly constant.
- **Controlled Environment:** Parameters like pH, temperature, oxygen, and nutrient supply can be precisely regulated.
- **Chemostat & Turbidostat:** Two common devices for continuous culture:
 - **Chemostat:** Growth rate is controlled by limiting nutrient supply.
 - **Turbidostat:** Growth rate is controlled by maintaining constant cell density.

Applications:

- Large-scale production of antibiotics, enzymes, and vitamins.
- Studying microbial growth kinetics and metabolism under constant conditions.
- Wastewater treatment and bioremediation studies.

(iii) Microbial leaching.?**Ans-Microbial Leaching (Bioleaching):**

Microbial leaching is the process of using microorganisms to extract metals from their ores. Certain bacteria and fungi can solubilize metals, making them easier to recover. This method is eco-friendly and cost-effective compared to traditional smelting.

Key Points:**1. Microorganisms Used:**

- *Acidithiobacillus ferrooxidans* (oxidizes iron and sulfur compounds)
- *Leptospirillum ferrooxidans* (iron-oxidizing bacteria)
- Fungi like *Aspergillus* for certain metals.

2. Mechanism:

- Microbes oxidize metal sulfides to soluble metal salts.
- Example: Copper sulfide → copper sulfate (soluble).

3. Applications:

- Extraction of copper, uranium, gold, and nickel from low-grade ores.
- Recovery of metals from electronic waste.

4. Advantages:

- Environmentally friendly (less air pollution).
- Economical for low-grade ores.

5. Limitations:

- Slow process.
- Only applicable to certain ores and metals.

(iv) Food chain.?

Ans- A food chain is a linear sequence that shows how energy and nutrients flow from one organism to another in an ecosystem. It begins with **producers** (plants) that make their own food through photosynthesis.

Herbivores (primary consumers) eat the producers, **carnivores** (secondary or tertiary consumers) eat the herbivores or other carnivores, and **decomposers** like fungi and bacteria break down dead organisms, returning nutrients to the soil.

Example:

Grass → Grasshopper → Frog → Snake → Hawk

Key points:

- Shows **energy transfer** from one level to the next.
- Usually **starts with plants** and ends with top predators.
- Helps maintain **ecosystem balance**.

(v) Biofertilizers.?

Ans- Biofertilizers are natural fertilizers containing living microorganisms that promote plant growth by increasing the availability of nutrients in the soil. They improve soil fertility and plant productivity in an eco-friendly way.

Types of Biofertilizers:

1. **Nitrogen-fixing bacteria** – e.g., *Rhizobium* (in legumes), *Azotobacter* (non-legumes)
2. **Phosphate-solubilizing bacteria** – e.g., *Bacillus*, *Pseudomonas*
3. **Potassium-solubilizing bacteria** – e.g., *Frateuria aurantia*
4. **Mycorrhizal fungi** – enhance nutrient and water uptake
5. **Cyanobacteria** – fix nitrogen in rice fields

Advantages:

- Environmentally safe, reduce chemical fertilizer use
- Enhance soil fertility and structure
- Promote sustainable agriculture

Limitations:

- Slower action compared to chemical fertilizers
- Effectiveness depends on soil and climate conditions

UNIT-I

2. what is Transcription. Explain how it transcripts in prokaryotes.?

Ans- Transcription-Transcription is the process by which the genetic information encoded in **DNA** is copied into **RNA**. It is the first step in gene expression and is essential for the synthesis of proteins. In prokaryotes, transcription produces **mRNA, tRNA, and rRNA**, which are directly involved in protein synthesis.

Key Features of Prokaryotic Transcription

1. Location:

- Occurs in the **cytoplasm** of prokaryotes because they **lack a nucleus**.

2. Enzyme Involved:

- **RNA polymerase** is the main enzyme that synthesizes RNA from the DNA template.
- Prokaryotic RNA polymerase consists of:
 - **Core enzyme:** α_2 , β , β' subunits (responsible for RNA synthesis)
 - **Sigma (σ) factor:** helps in recognizing the promoter region on DNA.

3. Template:

- Only **one strand** of DNA (the template strand) is used for transcription.
- The RNA is complementary to the **template strand** and identical (except U replaces T) to the **coding strand**.

Steps of Transcription in Prokaryotes

Transcription can be divided into **three main stages**:

1. Initiation

- **Promoter recognition:**
 - The σ factor of RNA polymerase recognizes and binds to the **promoter** sequence of DNA, typically the **-10 (TATAAT) and -35 (TTGACA) regions** upstream of the gene.
- **Formation of the transcription initiation complex:**
 - RNA polymerase binds to DNA, unwinds a small portion of it to expose the template strand.
- **Start of RNA synthesis:**
 - RNA polymerase initiates RNA synthesis at the **+1 site** (start site of transcription).

2. Elongation

- RNA polymerase moves along the DNA template in the **3' → 5' direction**.
- RNA is synthesized in the **5' → 3' direction**.
- **Complementary base pairing:**
 - A pairs with U, T with A, G with C, and C with G in RNA.
- The DNA helix **rewinds behind RNA polymerase** as the RNA transcript elongates.

3. Termination

Prokaryotic transcription ends when RNA polymerase encounters a **terminator sequence**. There are **two types of termination**:

1. **Rho-independent termination:**
 - Involves **GC-rich inverted repeats** in the RNA that form a **hairpin loop** followed by a series of **U residues**, causing RNA polymerase to detach.
2. **Rho-dependent termination:**
 - Requires the **Rho protein**, which binds to the RNA and moves toward RNA polymerase to **release the RNA transcript** at the termination site.

Post-transcription Events in Prokaryotes

- In prokaryotes, **mRNA is often polycistronic**, meaning it can code for multiple proteins.
- Unlike eukaryotes, **prokaryotic mRNA does not require splicing or capping**, and translation can occur **simultaneously with transcription**.

DNA → RNA → Protein (via Translation)

1. **Initiation** – RNA polymerase binds promoter, unwinds DNA.
2. **Elongation** – RNA polymerase synthesizes RNA complementary to DNA template.
3. **Termination** – RNA polymerase releases RNA at terminator.

UNIT-II

3. Explain microbial growth kinetics.?

Ans- Microbial growth kinetics is the study of the rate and pattern of microbial population increase over time under defined environmental and nutritional conditions. It is crucial in microbiology, biotechnology, fermentation technology, and food microbiology.

- Microorganisms grow by increasing **cell number**, not cell size.
- Growth can be described **quantitatively** using **kinetic models**, which are mathematical representations of microbial proliferation.
- Growth kinetics helps in:
 - Designing **fermentation processes**
 - Optimizing **production of antibiotics, enzymes, and biofuels**
 - Controlling **microbial contamination in food and pharmaceuticals**

2. Growth Curve

Microbial growth in a **batch culture** (closed system) typically follows a **sigmoidal pattern** with four phases:

1. Lag Phase

- **Characteristics:** No increase in cell number; cells metabolically active, adjusting to the new environment.
- **Reason:** Synthesis of enzymes, cofactors, ribosomes.
- **Kinetics:** Specific growth rate (μ) ≈ 0 .

2. Exponential (Log) Phase

- **Characteristics:** Cells divide at a constant and maximal rate.
- **Kinetics:**
 - Growth is **exponential**, described by:

$$N_t = N_0 e^{\mu t}$$

where:

- N_t = cell number at time t
- N_0 = initial cell number
- μ = specific growth rate (h^{-1})
- **Doubling time / generation time (g):**

$$g = \frac{\ln 2}{\mu}$$

- **Significance:** Most active metabolic phase; important for industrial production.

3. Stationary Phase

- **Characteristics:** Growth ceases; rate of cell death = rate of cell division.
- **Cause:** Nutrient depletion, accumulation of toxic metabolites.
- **Kinetics:** $\mu = 0$; total cell number constant.

4. Death (Decline) Phase

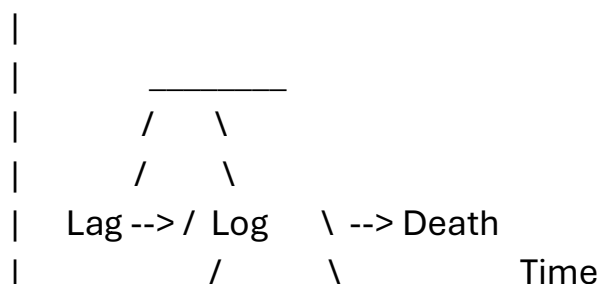
- **Characteristics:** Cell number decreases exponentially due to nutrient exhaustion or toxin accumulation.
- **Kinetics:** Can be modeled as:

$$N_t = N_s e^{-k_d t}$$

where k_d is the death rate constant.

Diagram: Microbial growth curve

Cell number



3. Factors Affecting Microbial Growth Kinetics

1. Nutrient availability

- Limiting substrates slow growth (Monod kinetics).
- Monod equation:

$$\mu = \mu_{max} \frac{S}{K_s + S}$$

Where:

- S = substrate concentration
- μ_{max} = maximum specific growth rate
- K_s = half-saturation constant

2. Environmental conditions

- Temperature, pH, oxygen, osmotic pressure influence μ and g .

3. Waste accumulation

- By-products can inhibit growth (toxic metabolites).

4. Types of Growth Kinetics

1. Batch culture kinetics

- Closed system, nutrient-limited.
- Exhibits all 4 growth phases.

2. Continuous culture (chemostat) kinetics

- Fresh medium continuously added, culture withdrawn at same rate.
- Maintains **steady-state growth**.
- Growth rate controlled by **dilution rate (D)**:

$$\mu = D$$

3. Fed-batch kinetics

- Substrate added periodically to prolong exponential phase.
- Common in industrial fermentation.

5. Mathematical Models

1. Exponential growth model

$$N_t = N_0 e^{\mu t}$$

2. Logarithmic form

$$\ln N_t = \ln N_0 + \mu t$$

3. Monod model (substrate-limited growth)

$$\mu = \mu_{max} \frac{S}{K_s + S}$$

4. Logistic growth model

$$\frac{dN}{dt} = \mu N \left(1 - \frac{N}{N_{max}} \right)$$

6. Measurement of Microbial Growth

1. Direct methods

- Viable cell count (plate count method)
- Microscopic count (hemocytometer)

2. Indirect methods

- Turbidity (optical density, OD)
- Biomass dry weight
- Metabolic activity (CO₂ evolution, substrate consumption)

7. Applications

- **Industrial microbiology:** Optimizing fermentation of antibiotics, enzymes, organic acids.
- **Food microbiology:** Predicting shelf life and microbial spoilage.
- **Environmental microbiology:** Bioremediation kinetics.
- **Pharmaceuticals:** Standardizing microbial inocula.

8. Conclusion

Microbial growth kinetics is a fundamental concept in microbiology, describing how microorganisms proliferate under different conditions. It combines **experimental measurement** with **mathematical modeling**, enabling prediction and optimization in **industrial, environmental, and clinical settings**.

Understanding growth kinetics ensures **maximum yield, efficient resource use, and control over unwanted microbial growth**.

UNIT-III

4.What is replication? How it takes place in Prokaryotes.?

Ans- Replication: Definition

Replication is the biological process by which a cell makes an **exact copy of its DNA** before cell division. It ensures that each daughter cell receives a complete set of genetic information identical to the parent cell.

Key features of replication:

- **Semi-conservative:** Each new DNA molecule contains one original (parental) strand and one newly synthesized strand.
- **Accurate:** DNA polymerases proofread to minimize errors.
- **Bidirectional:** Replication proceeds in two opposite directions from the origin of replication (in prokaryotes).

DNA Replication in Prokaryotes

Prokaryotic DNA is generally **circular** and **double-stranded**, e.g., in *Escherichia coli*. Replication in prokaryotes occurs in **four main stages**:

1. Initiation

- Replication begins at a specific site called the **origin of replication (OriC)**.
- **Initiator protein (DnaA)** binds to OriC, causing the DNA to unwind.
- **Helicase (DnaB)** further unwinds the DNA strands, forming a **replication fork**.

- **Single-strand binding proteins (SSBs)** stabilize the separated strands to prevent them from reannealing.
- **Topoisomerase (DNA gyrase)** relieves the supercoiling ahead of the replication fork.

2. Elongation

- **Primase** synthesizes short RNA primers complementary to the DNA strand.
- **DNA polymerase III** adds nucleotides in the **5' → 3' direction**.
- Because DNA strands are **antiparallel**:
 - **Leading strand**: Synthesized continuously toward the replication fork.
 - **Lagging strand**: Synthesized discontinuously away from the fork as **Okazaki fragments**.
- **DNA polymerase I** removes RNA primers and replaces them with DNA.
- **DNA ligase** seals the gaps between Okazaki fragments, forming a continuous strand.

3. Termination

- Replication continues until the forks meet at the **termination site (Ter)**.
- **Tus proteins** bind to Ter sequences and stop the movement of helicase.
- Circular DNA molecules are **decatenated (unlinked)** by **topoisomerase IV**, producing two identical DNA molecules.

4. Proofreading and Accuracy

- DNA polymerase III has **3' → 5' exonuclease activity**, allowing it to remove incorrectly paired nucleotides.
- This ensures high fidelity of DNA replication.

Summary / Key Points

Feature	Prokaryotic DNA Replication
DNA Type	Circular, double-stranded
Origin	Single origin (OriC)
Direction	Bidirectional
Polymerase	DNA pol III (major), DNA pol I (primer removal)
Strand Synthesis	Leading (continuous), Lagging (Okazaki fragments)
Proofreading	DNA pol III exonuclease activity

Diagram (simplified prokaryotic replication fork)

Replication Fork
 ↓
 Leading strand → -----
 Lagging strand ← ----O----O----O--- (Okazaki fragments)
 RNA primers (•)

Important Notes for 45 Marks:

- Mention **semi-conservative and bidirectional nature**.
- Include **enzymes involved** (helicase, primase, DNA polymerase, ligase, gyrase).
- Explain **leading and lagging strand synthesis** clearly.
- Add **termination and proofreading**.
- A labeled **diagram of replication fork** is highly recommended.

UNIT-IV

5. What are Biofertilizers.?

Ans- Biofertilizers are microbial inoculants containing living microorganisms which, when applied to seeds, plant surfaces, or soil, **colonize the rhizosphere or the interior of the plant** and promote growth by increasing the supply or availability of primary nutrients to the host plant.

In simpler terms, biofertilizers are natural fertilizers made from **beneficial microbes** that help plants grow by improving nutrient availability.

Characteristics of Biofertilizers

1. **Microbial-based** – They contain bacteria, fungi, or algae that are beneficial to plants.
2. **Eco-friendly** – They are natural and do not cause pollution.
3. **Cost-effective** – Cheaper than chemical fertilizers.
4. **Soil fertility enhancer** – Improve soil health by enriching it with essential nutrients.
5. **Specific action** – Each biofertilizer has a specific function, e.g., nitrogen fixation, phosphate solubilization.

Types of Biofertilizers

1. Nitrogen-fixing Biofertilizers

- Convert atmospheric nitrogen into a form plants can use.
- Examples:
 - **Rhizobium** – Symbiotic bacteria for legumes (e.g., pea, soybean)
 - **Azotobacter** – Free-living nitrogen-fixing bacteria for cereals and vegetables
 - **Azospirillum** – Associative nitrogen-fixing bacteria for sugarcane, maize

2. Phosphate Solubilizing Biofertilizers

- Convert insoluble phosphate into soluble forms for plant uptake.
- Examples: **Bacillus**, **Pseudomonas**, **Aspergillus**

3. Potassium Solubilizing Biofertilizers

- Solubilize potassium from insoluble sources.
- Example: **Frateuria aurantia**

4. Vesicular-Arbuscular Mycorrhiza (VAM)

- Symbiotic fungi that enhance nutrient and water uptake, especially phosphorus.
- Example: **Glomus spp.**

5. Cyanobacteria (Blue-Green Algae)

- Photosynthetic microorganisms that fix atmospheric nitrogen.
- Commonly used in **paddy fields**.
- Examples: **Anabaena, Nostoc, Oscillatoria**

Mechanism of Action

1. **Nitrogen Fixation:** Convert atmospheric $N_2 \rightarrow NH_3 \rightarrow$ usable nitrogen for plants.
2. **Phosphate Solubilization:** Secrete organic acids that make phosphates soluble.
3. **Growth Promotion:** Some microbes produce **plant growth hormones** like auxins, gibberellins, and cytokinins.
4. **Soil Health Improvement:** Increase microbial population and soil organic matter.

Advantages of Biofertilizers

1. Enhance plant growth and yield naturally.
2. Reduce dependence on chemical fertilizers.
3. Eco-friendly and sustainable.
4. Improve soil fertility and structure.
5. Cost-effective for farmers.
6. Help in nitrogen, phosphorus, and potassium availability.

Limitations

1. Sensitive to **environmental conditions** (temperature, pH, moisture).
2. Short shelf-life – microbes lose viability if not stored properly.
3. **Slow action** compared to chemical fertilizers.
4. Requires **proper application techniques** for effectiveness.

Applications

- Used in **seed treatment**, soil application, or root dipping.
- Common crops: legumes, cereals, vegetables, sugarcane, paddy, and horticultural plants.
- Often combined with **organic fertilizers** for better results.

Conclusion

Biofertilizers are **natural, sustainable, and eco-friendly alternatives** to chemical fertilizers. They not only supply essential nutrients to plants but also improve soil fertility, reduce environmental pollution, and enhance crop productivity in a long-term and sustainable manner.